A NEW ERA OF BRITISH AVIATION



BRITISH AIRWAYS / PROJECT BRITANNIA

An imagined future for a faster, proudly British global airline.

A strategic concept brochure. Aircraft, performance figures and programme details are illustrative; they are not current British Airways or BAE Systems products.

01 / THE VISION

Time is the ultimate luxury.

Imagine a British Airways that does more than connect the world. It transforms the meaning of distance. A business meeting in New York before lunch; an evening in Rome after breakfast in London; a global network built around giving people more of their day back.

The Mach Generation is a concept for a new family of British-designed high-speed passenger aircraft, developed through a proposed industrial consortium led by BAE Systems, with Rolls-Royce and British manufacturing partners. British Airways would act as launch customer and operator.

The ambition is not to make every flight supersonic at any cost. It is to introduce high speed where it creates meaningful value, while keeping safety, comfort, reliability and commercial discipline at the centre.

THREE PRINCIPLES

FLY FASTER Build a premium network around routes where reduced journey time matters most.	**BUY BRITISH** Develop domestic capability in design, propulsion, materials and maintenance.	**MAKE IT WORK** Match ambition with certification, environmental performance and sound economics.

THE BRITISH INDUSTRIAL IDEA

Project Britannia would seek to link aircraft design, propulsion research, advanced materials, precision manufacturing, digital flight systems and lifecycle maintenance into one coordinated British aerospace programme. The commercial test is clear: the aircraft must be safe, certifiable, desirable and profitable.

The lesson of Concorde is both inspiring and practical. British and French engineering achieved extraordinary supersonic performance, but technical brilliance alone did not guarantee commercial viability. The next generation must pair ambition with better economics.

02 / THE FLEET

A family of British aircraft.

The proposed fleet is a design brief, not a catalogue of existing aircraft. Each variant would be engineered for a distinct market, with speed balanced against range, capacity, noise and operating cost.

AIRCRAFT	ROLE	TARGET CRUISE	PROPOSED CAPACITY	INDICATIVE RANGE
Britannia S1	Premium regional / shorter international	Up to Mach 1.2	40–60	3,000 km
Britannia S2	Transatlantic flagship	Mach 1.6–1.8	80–120	7,000 km
Britannia L1	Long-range international	Mach 1.6–2.0	100–150	10,000 km



DESIGN DIRECTION / A recognisable British Airways identity: white fuselage, midnight-blue lower body, restrained red detailing and a distinctive Union Flag-inspired tail.

All specifications are illustrative targets. They would require engineering validation, flight testing, certification and a viable operating case. A Mach 1 aircraft cannot automatically operate supersonically over every land route.

03 / THE NETWORK

Europe, brought closer.

At a reference speed of 1,070 km/h—approximately Mach 1 under typical high-altitude conditions—the distance between major cities looks very different. These are simplified airborne equivalents based on approximate great-circle distances, not published timetables.

DESTINATION	APPROX. DISTANCE	MACH 1 EQUIVALENT
Paris	345 km	19 min
Amsterdam	360 km	20 min
Brussels	320 km	18 min
Frankfurt	635 km	36 min
Zurich	780 km	44 min
Milan	955 km	54 min
Rome	1,435 km	1 hr 20 min
Barcelona	1,135 km	1 hr 4 min
Madrid	1,265 km	1 hr 11 min
Copenhagen	955 km	54 min
Stockholm	1,435 km	1 hr 20 min
Athens	2,395 km	2 hr 14 min

The reality check: on very short journeys, airport access, security, boarding and acceleration can outweigh cruise-speed savings. Conventional aircraft and high-speed rail would remain highly competitive on many European routes. Supersonic operations would also be subject to applicable overland speed and noise rules.

04 / THE ATLANTIC & BEYOND

Make the long journey shorter.

The North Atlantic is the natural first market to investigate: established premium demand, substantial business travel and a clear value proposition for time saved. Longer routes to the Middle East and Asia could follow only where range, payload, reserves and operating approvals support them.

DESTINATION	APPROX. DISTANCE	MACH 1 EQUIVALENT
New York	5,570 km	5 hr 12 min
Boston	5,250 km	4 hr 54 min
Washington, D.C.	5,890 km	5 hr 30 min
Chicago	6,350 km	5 hr 56 min
Toronto	5,700 km	5 hr 20 min
Dubai	5,500 km	5 hr 9 min
Doha	5,200 km	4 hr 52 min
Mumbai	7,200 km	6 hr 44 min
Delhi	6,720 km	6 hr 17 min
Hong Kong	9,620 km	8 hr 59 min
Tokyo	9,560 km	8 hr 56 min
Singapore	10,880 km	10 hr 10 min
Sydney	16,990 km	15 hr 53 min

These figures are distance ÷ 1,070 km/h. They exclude climb, acceleration, descent, winds, routing, holding and ground time. A Mach 1.8 cruise would reduce the theoretical cruise-time equivalent, but actual schedules would be longer and would depend on aircraft performance and operating restrictions.

A route is a candidate, not a promise: Sydney, for example, lies beyond the illustrative 10,000 km L1 range on a great-circle basis and would require a different aircraft, a stop, or another operating concept.

05 / THE EXPERIENCE

British elegance at high speed.

The cabin should feel unmistakably British Airways: considered, calm and premium. Natural materials, midnight-blue upholstery, brushed metal, refined lighting and acoustic engineering would make speed feel effortless rather than theatrical.

On the flagship aircraft, a premium cabin could offer private suites, lie-flat seating and dedicated dining. A business-focused regional cabin would prioritise personal space, privacy and productivity. The wider network would connect supersonic and conventional flights in one booking.

TIME ADVANTAGE

Before departure Priority processing and dedicated premium check-in.	**In the air** A quieter, carefully designed cabin with space to work or rest.
On arrival Reliable baggage handling and well-timed onward connections.	**Across the network** Door-to-door journey planning, not just a faster cruise.

06 / BUY BRITISH

An industrial programme, not just an aircraft order.

Project Britannia would seek to retain as much high-value design, manufacturing and lifecycle support in Britain as commercially and technically feasible.

CAPABILITY	PROPOSED CONTRIBUTION
BAE Systems	Airframe, aerodynamic integration and systems engineering
Rolls-Royce	Propulsion research, engines and technical support
British suppliers	Materials, precision components, cabin systems and electronics
Universities & skills	Research partnerships, simulation and workforce development
UK service centres	Maintenance, repair, upgrades and lifecycle support

Potential industrial regions for assessment include Lancashire, Derby, Bristol and the South West, South Wales, and major airport maintenance hubs. These are candidates for study, not confirmed project sites. The programme should use long-term supplier agreements and independent cost reviews to make British content sustainable.

07 / THE BUSINESS CASE

Speed must earn its place.

The project should begin with a small fleet on routes where passengers are willing to pay for meaningful time savings. It should complement—not immediately replace—the conventional long-haul fleet.

PHASE	OBJECTIVE	GATE
1. Feasibility	Route economics, passenger research and aircraft requirements	Approve research funding
2. Technology	Aerodynamics, propulsion, materials and noise studies	Validate demonstrator results
3. Prototype	Build and test a development aircraft	Independent technical review
4. Certification	Demonstrate airworthiness and environmental compliance	Production decision
5. Initial service	Operate a limited commercial fleet	Prove reliability and profitability
6. Expansion	Increase production and broaden the network	Scale only when justified

Five tests should guide investment: revenue per flight, load factor, aircraft utilisation, maintenance economics and capital discipline. Development expenditure, unit production cost and airline operating costs must be modelled separately. Precise ROI would be speculative until these inputs are established.

08 / RESPONSIBLE HIGH SPEED

Designed for the realities of tomorrow.

Supersonic flight brings particular challenges: aerodynamic drag, fuel consumption, high-altitude emissions, airport noise and sonic booms. The programme must address these from the outset rather than treating them as later-stage details.

Design priorities should include lower drag and structural weight, efficient propulsion, compatibility with sustainable aviation fuels where feasible, applicable noise and emissions certification, low-boom research, and transparent reporting of fuel use and emissions per passenger-kilometre.

Conventional aircraft and rail will remain the right answer for many journeys. The aim is maximum useful speed where it can be delivered safely, legally and responsibly.

THE FUTURE IS A LITTLE CLOSER.

BRITISH AIRWAYS
THE MACH GENERATION

Buy British. Fly faster. Make the world closer.



A concept for a future British aerospace and airline programme. Project Britannia, its aircraft, specifications, performance targets and industrial structure are fictional illustrations, not an announcement of an existing British Airways or BAE Systems programme.

Route times are theoretical distance/speed calculations at a reference 1,070 km/h. They are not scheduled flight times, and actual flight times would be longer. Aircraft design, performance, route permissions, certification, environmental effects and commercial viability would all require independent validation.